Cloud Computing

Exercise - 5 Total points: 10

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1. In Exercise – 4, Task 2, you wrote a multithreaded **Java** program to download online books from Gutenberg repository.
   Now you have learnt the Java Sockets. With Java Sockets, you can send messages to other processes on same/different machines.
   a. Now upgrade your master/main program to download the files by taking the help of multiple processes.
      Hint: Send a message to trigger a threaded program on each machine. The program downloads a subset of files provided by the master node. (2 points)
      Note: Provide a sequence diagram of your protocol along with messages being exchanged between the processes.
   b. Now assume that all the machines are downloading the files and storing them in a shared repository. Achieve it within/outside the java program by strictly using scp command, and copy the files to your lab server, within your user space, into a Gutenberg subfolder. (2 points)
   c. Now extend the program to also have the search facility in the downloaded books. (2 points)
      Hint: Now you have to send a response back to master node and the master node has to consolidate all the inputs from different machines.
      Note: Think of all the cases such as what will happen when there are network failures? What is your fall-back mechanism.
      Note: Provide a sequence diagram of your protocol along with messages being exchanged between the processes.

2. Calculate the Monte Carlo PI by using similar approach of taking help from multiple processes on different machines. (2 points)
   Note: You still need to use the multithreaded program on the slave nodes.

3. Until now you have fixed the machines which participate in this distributed computing setup. Extend the scenario where several slaves can approach the master and register themselves before. The master maintains a list of all the slaves and then uses them when a job such as Monte Carlo PI is initiated. Demonstrate the scenario with Monte Carlo PI. (2 points)
   Note: Do not forget the importance of exchanging the IP addresses and port numbers of the slaves with the master.

<u>Follow these guidelines while submitting the exercises</u>

1. Generate a PDF file with answers/screenshots with name being your studentid_EX5
2. Export the eclipse project (for java code) as a zip file
3. Put a pdf file with the answers for the questions
   a. For programming tasks show screenshots of the program output for the correct and wrong inputs
   b. For commands tried, provide the screenshots
4. The programs should guide the user in putting the correct input

    a. Have detailed comments of your logic
5. Do not copy code/answers from internet. If you copy, please leave a comment with the URL on that java file
6. Plagiarism will not be tolerated and dealt through the formal channels of the university